Sreekaran Reddy Ramasahayam

(913) 299-7216 | sreekaran.2021@gmail.com | linkedin.com/in/sree1704 | github.com/Sreekaran1704 | sreekaran-portfolio.vercel.app

Professional Summary

Data Scientist with an M.S. in Computer Science (Data Science Track, GPA 3.97), specializing in causal inference, predictive modeling, and applied AI systems, with additional depth across analytics engineering, BI dashboard architecture, MLOps deployment, and business analysis. Builds end-to-end pipelines from raw data through validated statistical models and deployed AI applications, with a track record spanning a validated $3.73–$5.00 per-customer revenue lift ($p < 0.0001$), a 30% gain in reporting reliability, and production-style ML systems deployed on Kubernetes.

Technical Skills

Machine Learning and Statistics: scikit-learn, XGBoost, Causal Inference (RCT, Propensity Score Matching, Difference-in-Differences), Regression, Classification, A/B Testing, Hypothesis Testing, Feature Engineering, Cross-Validation, Model Evaluation

Applied AI and LLMs: Hugging Face, PyTorch, LoRA, QLoRA, RAG, FAISS, Prompt Engineering, LLM-as-Judge, Model Context Protocol (MCP), Groq API

MLOps and Deployment: FastAPI, Docker, Kubernetes, AWS EKS/EC2/S3, GitHub Actions, CI/CD, MLflow, DVC, Prometheus/Grafana

Analytics Engineering and BI: dbt Core, Semantic Modeling, Schema Design, Tableau (Calculated Fields, Parameters, LOD Expressions), Dashboard Architecture, Excel

Data Engineering and Warehousing: SQL, Python, Pandas, NumPy, statsmodels, ETL Workflows, BigQuery, PostgreSQL, DuckDB, MongoDB, Google Cloud SQL, Git

Business Analysis: Requirements Gathering, Stakeholder Communication, Agile/Scrum, JIRA, Confluence, KPI Definition, Process Improvement

Professional Experience

Sree Nirman May 2023 – Jun 2024

Data Analyst and Machine Learning Engineer

- Owned data quality across a 50K+ record construction portfolio spanning cost, labor, budget, and sales, standardizing fragmented schemas using SQL and Pandas to lift reporting reliability by 30%.
- Designed SQL logic using joins, CTEs, CASE expressions, and window functions to standardize cost, labor, and sales calculations across variance, productivity, and progress-tracking metrics.
- Addressed a limited sample of 150–200 real project records by generating synthetic training records sampled from real feature distributions, scaling the dataset to a size viable for reliable model benchmarking.
- Engineered 20+ predictive features across 7+ project dimensions, benchmarking Ridge and Lasso regression against a baseline linear model using R^2, MSE, and MAE to estimate bid pricing within 10–20% of held-out results.
- Tracked and compared model experiments using MLflow, logging runs and metrics across candidate models and registering the best-performing version for reuse.
- Delivered Tableau dashboards, Excel scorecards, and executive reporting packs presenting bi-weekly to stakeholders, driving a 15% gain in operational efficiency.
- Configured a scheduled Tableau Server extract refresh, automating weekly dashboard updates and eliminating manual republishing.
- Tracked reporting and analysis work as JIRA tickets within two-week sprints, coordinating delivery through an Agile/Scrum workflow.

Avanthi High School Apr 2022 – Jan 2023

Data Analyst Intern

- Built the school's analytics foundation from scratch, structuring 12K+ student financial records and 50K+ expense records into validated, reporting-ready datasets using SQL, Python, and Tableau.
- Analyzed 50K+ expense transactions across dining, hostel, and academics, identifying 10–15% in cost-saving opportunities that supported the institution's first break-even cycle within four months.
- Tuned an XGBoost classifier on engineered academic features via grid search and cross-validation, predicting 10th-grade performance from prior records with SHAP-based feature importance validating model behavior.
- Built an OCR pipeline to extract text from scanned admission answer sheets, then fine-tuned GPT-2 on rubric-conditioned records to score descriptive responses, replacing manual grading.
- Validated model-driven scholarship outputs against 1K+ principal-approved records before rollout, informing a merit-based scoring redesign that replaced an inconsistent allocation process.
- Designed a two-channel A/B comparison isolating the admission test's marketing impact, driving an 18% rise in admissions.
- Launched admission-test fees as a new revenue stream alongside the scoring redesign, contributing to a 50% rise in school revenue.

University of Missouri–Kansas City Aug 2025 – May 2026

Information Services Lab Assistant

- Own technical support for student-facing computer labs, resolving 75+ tickets per semester across hardware, printer, login, and workstation issues, alongside Python, SQL, R, and notebook-based coding support.
- Drive analytics coaching across 20+ student academic and research projects in Python and R, strengthening EDA rigor, statistical validation, and assumption-checking in peer analysis.

Projects

FanHouse: Membership Program Causal Impact Study | *Python, scikit-learn, SciPy* GitHub · Live

– Designed and ran a randomized experiment across 50,000 online customers and 325 stores, isolating a statistically significant revenue lift of \$3.73–\$5.00 per customer per quarter ($p < 0.0001$).
– Demonstrated an 8x overstatement in a naive self-selected comparison using a placebo test, then corrected it with propensity score matching and difference-in-differences, two independent methods converging on the same estimate.
– Scaled the validated per-customer effect to FanHouse's full footprint, projecting an annualized net revenue uplift of \$11.6M–\$19.9M net of discounts and coin liabilities.

Market Pulse: Job Market Intelligence Platform | *FastMCP, FastAPI, PostgreSQL, Docker, Groq API* GitHub · Live

– Refactored shared query logic into a standalone module and built a dual-transport MCP server on top of it, exposing five job-market analytics tools over local stdio and remote streamable HTTP mounted on an existing FastAPI app.
– Secured the remote HTTP transport with bearer-token authentication and a per-IP sliding-window rate limiter, engineering a lifespan-managed session handler to keep the async context alive within FastAPI's startup lifecycle.
– Automated daily ingestion of live postings from the Adzuna and RemoteOK APIs into a normalized PostgreSQL schema via SQLAlchemy, pre-aggregating daily snapshots for structured querying.
– Layered a Groq-powered LLM summary on top of the computed statistics, translating raw hiring trends into a plain-English weekly market report grounded strictly in the underlying data, with no hallucinated figures.
– Deployed the full service on Render using Docker and GitHub Actions, automating the ingestion-to-delivery cycle to run continuously without manual intervention.

MedPredict-X: Readmission Forecasting with RAG Reasoning | *XGBoost, FAISS, Qwen-32B* GitHub

– Built a hospital readmission forecasting system on 100K+ patient encounters, comparing five model types and selecting XGBoost to balance precision and recall for 30-day readmission risk.
– Designed probability-based risk tiers, routing high-risk patients to a doctor call within 2–5 days, moderate-risk to a nurse call, and low-risk to automated outreach, translating model scores into concrete next actions.
– Built a RAG explanation layer using FAISS retrieval and Qwen-32B to generate human-readable reasoning tied to each patient's clinical and contextual factors.

Vehicle Insurance Eligibility MLOps Pipeline | *FastAPI, Docker, Kubernetes, AWS EKS, GitHub Actions*

– Built a classification pipeline predicting insurance eligibility from customer, vehicle, and claims history, served through a FastAPI inference endpoint with DVC-based data and model versioning for reproducibility.
– Containerized the service with Docker and deployed it as pods and deployments on AWS EKS, automating CI/CD-style updates with GitHub Actions for production-style model serving.
– Monitored model accuracy and drift over time in production using Prometheus and Grafana, enabling proactive detection of degrading predictions post-deployment.

NorthMart: Retail Analytics Pipeline | *SQL, dbt Core, BigQuery, Tableau*

– Designed and built a DuckDB-to-BigQuery analytics pipeline for a simulated 50-store, 5-region retail chain, transforming 15M+ raw transaction line items into analysis-ready fact and dimension tables.
– Modeled the warehouse as a star schema, separating a central sales fact table from store, product, and date dimensions, and implemented dbt Core models with data-quality tests and lineage documentation.
– Published Tableau Public dashboards translating 15M+ transaction records into store- and region-level performance views, structuring the schema around promotions, supplier trends, and inventory risk.

Multi-Object Tracking: Pedestrian Detection and Re-Identification | *PyTorch, Faster R-CNN, OpenCV*

– Fine-tuned a Faster R-CNN detector (ResNet-50 backbone) and trained a Siamese re-identification network (ResNet-18 backbone, contrastive loss) on the MOT16 benchmark, as part of a 4-person team.
– Improved re-identification accuracy from 91.2% to 97.2%, with precision reaching 0.948 on the held-out benchmark split.

Education

University of Missouri–Kansas City Aug 2024 – May 2026
Master of Science in Computer Science, Data Science Track *GPA: 3.97*

– Relevant Coursework: Database Management Systems, Statistical Learning, Principles of Data Science, Cloud Computing, Generative AI

Certifications

Salesforce Certified AI Associate